Deutsch-Jozsa Algorithm

→ Deutsch-Jozsa Problem

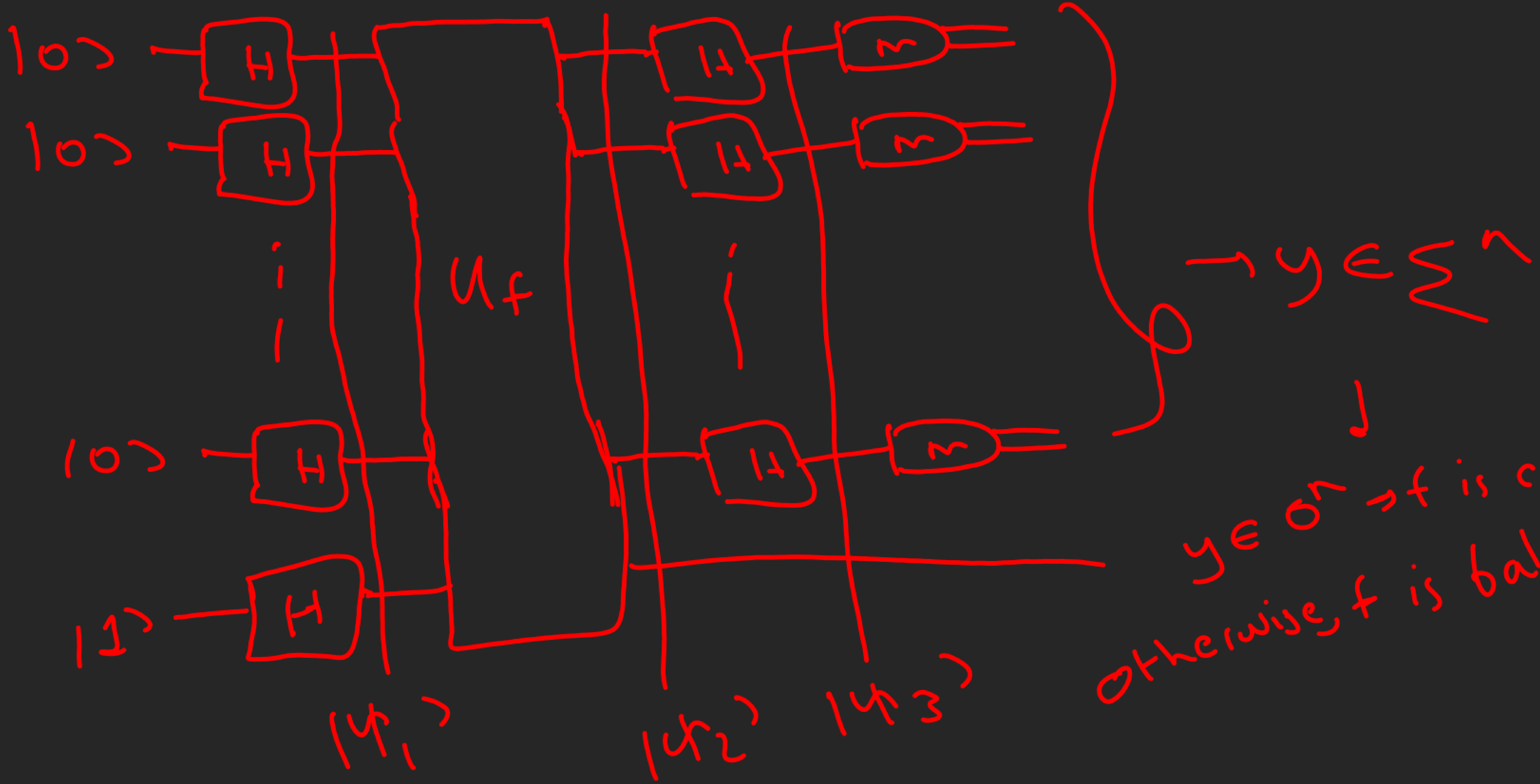
Input → $f : \Sigma^n \rightarrow \Sigma$

Promise → f is either constant or balanced

Output → 0 if f is constant
→ 1 if f is balanced

a	$f(a)$
00	0
01	0
10	0
11	1

→ Niether constant, nor balanced
↓
We ignore it



$y \in 0^n \to f$ is constant

otherwise, f is balanced

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \qquad H|a\rangle = \frac{|0\rangle + (-1)^a |1\rangle}{\sqrt{2}}$$

$$H^{\otimes n} = \underbrace{H \otimes H \_\_\_\_ \otimes H}_{n \text{ times}}$$

$$\Rightarrow H|a\rangle = \frac{1}{\sqrt{2}} \sum_{b \in \{0,1\}} (-1)^{ab} |b\rangle$$

$$H^{\otimes n} |x_0, x_1 \_\_\_\_ x_{n-1}\rangle = (H|x_0\rangle) \otimes (H|x_1\rangle) \_\_\_\_ \otimes (H|x_{n-1}\rangle)$$

$$= \left(\frac{1}{\sqrt{2}} \left(\sum_{y_{n-1} \in \{} (-1)^{x_{n-1} \cdot y_{n-1}} |y_{n-1}\rangle \right) \right) \otimes \_\_\_\_ \left(\frac{1}{\sqrt{2}} \left(\sum_{y_0 \in \{} (-1)^{x_0 \cdot y_0} |y_0\rangle \right) \right)$$

$$= \left(\frac{1}{\sqrt{2^n}} \sum_{y_{n-1} - y_0 \in \{} (-1)^{x_{n-1} \cdot y_{n-1} \oplus \_\_\_\_ \oplus x_0 y_0} |y_{n-1} \_\_\_\_ y_0\rangle \right.$$

$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle$ → Remember

Binary Dot Product → $x \cdot y = x_{n-1} \cdot y_{n-1} \oplus \_\_ \oplus x_0 y_0$

$$= \begin{cases} 0 & \text{if } x_{n-1} y_{n-1} \_\_ x_0 y_0 \text{ is even} \\ 1 & \text{if } x_{n-1} \cdot y_{n-1} \_\_ x_0 y_0 \text{ is odd} \end{cases}$$

$|\psi_1\rangle = H^{\otimes n}|0\rangle \otimes H|1\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} |y\rangle \otimes |-\rangle = |\epsilon\rangle \otimes |-\rangle$

$|\psi_2\rangle$

$U_f(|a\rangle|-\rangle) = (-1)^{f(a)}|a\rangle|-\rangle$

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle |-\rangle$$

→ Phase kickback

→ Replace dummy variable, y with x

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle$$

measured

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} \sum_{y \in \{0,1\}^n} (-1)^{f(x) + x \cdot y} |y\rangle \otimes |-\rangle$$

$$p(0^n) = \left| \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \right|^2$$

$P(0^n) = 1 \rightarrow$ when f is constant
$P(0^n) \neq 1 \rightarrow$ when f is balanced

$\rightarrow$ Any deterministic algorithm takes atleast $2^{n-1} + 1$ queries

$\rightarrow$ Probabililistic Algorithm

Choose k input strings $x_1, \_\_\_ x_k \in \Sigma^n$ then f is constant,
If $f(x_1) \_\_\_ = f(x_k)$ orelse f is balanced

If f is constant algorithm is correct with probability

1

If f is balanced algorithm is correct with probability

$1 - 2^{1-k}$